

L14:

Shrinkage methods

- Ridge regression
- LASSO

Recall: So far, we have studied empirical risk minimization (ERM) in the context of linear regression:

$$\min_{\beta \in \mathbb{R}^p} \|Y - X\beta\|^2$$



$$\min_{\beta \in \mathbb{R}^p} \sum_1^n (y_i - \langle \beta, x_i \rangle)^2$$

Note: We have not considered an intercept here; this can be achieved by centering the data as part of pre-processing:

$$\bar{x} \leftarrow \frac{1}{n} \sum_1^n x_i ; \quad \bar{y} = \frac{1}{n} \sum_1^n y_i$$

$$x_i \leftarrow x_i - \bar{x} ; \quad y_i \leftarrow y_i - \bar{y}$$

This eliminates the need for an intercept.
★ Assume in the remainder of this class that data is centered.

Typically, we prefer to add a regularizer into the ERM objective.

$$\min_{\beta \in \mathbb{R}^p} \|Y - X\beta\|^2 + \underbrace{\lambda \|\beta\|^2}_{\substack{\text{regularization} \\ \text{parameter} \\ \text{L}_2 \text{ regularizer}}}$$

RIDGE REGRESSION (RR)

- The regularization imposes a ~~pen~~ penalty to "large" regression fits β

- Tends to prevent overfitting

- But introduces bias

→ bias-variance tradeoff

- Regularization parameter λ is a hyperparameter

The solution of the ~~reg~~ ridge regression optimization is :

$$\hat{\beta}^{RR} = (\lambda I + X^T X)^{-1} X^T Y. \quad - (1)$$

- Compare with our earlier OLS solution (with $\lambda=0$)

$$\hat{\beta}^{OLS} = (X^T X)^{-1} X^T Y. \quad - (2)$$

② agrees with ① setting $\lambda=0$.

But ② only holds when X has full col. rank, i.e., $X^T X$ is invertible!

However, ① holds regardless of the col. rank of X .

{ This was in fact the original motivation for introducing the quadratic regularization - do away with the issues arising from $X^T X$ being poorly conditioned. }

In fact, RR is equivalent to the following ^{constrained} optimization:

$$\begin{array}{ll} \min_{\beta \in \mathbb{R}^p} & \|Y - X\beta\|^2 \\ \text{s.t.} & \|\beta\|^2 \leq t \end{array}$$

(3)

Indeed, the ~~lagrangian~~ Lagrangian for the above is:

$$\mathcal{L}(\beta, \alpha) = \|Y - X\beta\|^2 + \alpha (\|\beta\|^2 - t)$$

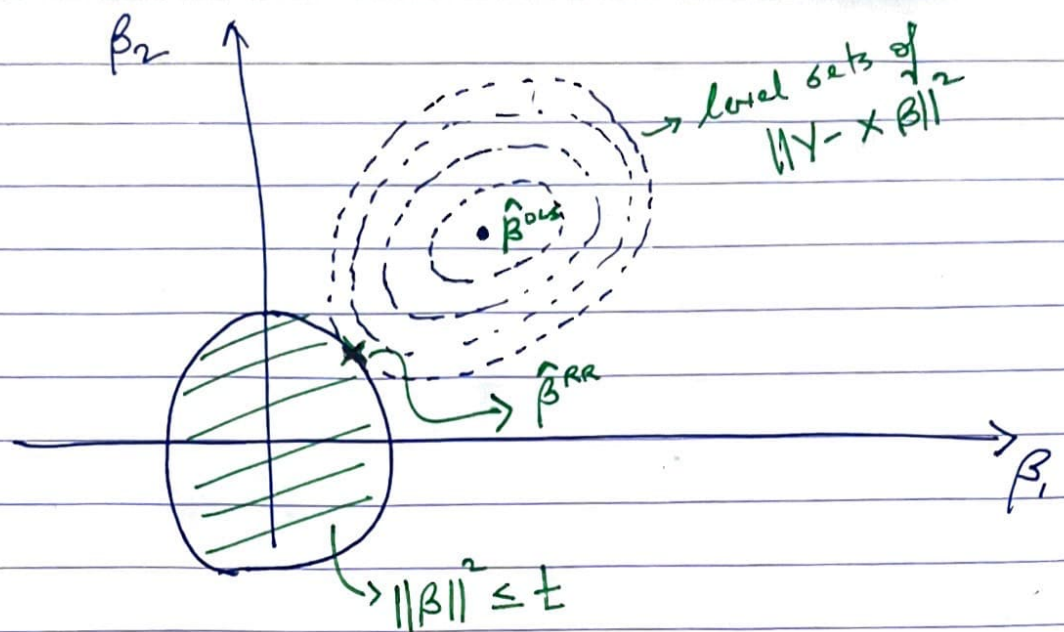
By KKT, β^* (solution of (3)) satisfies

$$\begin{aligned} \beta^* &= \arg \min_{\beta} \mathcal{L}(\beta, \alpha^*) \\ &= \arg \min_{\beta} \|Y - X\beta\|^2 + \alpha^* (\|\beta\|^2 - t) \end{aligned}$$

$$= \arg \min_{\beta} \|Y - X\beta\|^2 + \alpha^* \|\beta\|^2$$

★ Think of λ in RR as optimal Lagrange multiplier α^* of (3)

★ RR effectively solves (3) for some t , which depends on λ chosen.



Lasso

↳ Least absolute shrinkage & selection operator

Here, the ℓ_2 regularizer in RR is replaced by an ℓ_1 regularizer.

$$\min_{\beta \in \mathbb{R}^p} \|Y - X\beta\|^2 + \lambda \|\beta\|_1$$

LASSO

★ Here, $\|\beta\|_1 = \sum_{i=1}^p |\beta_i|$ (ℓ_1 norm of β)

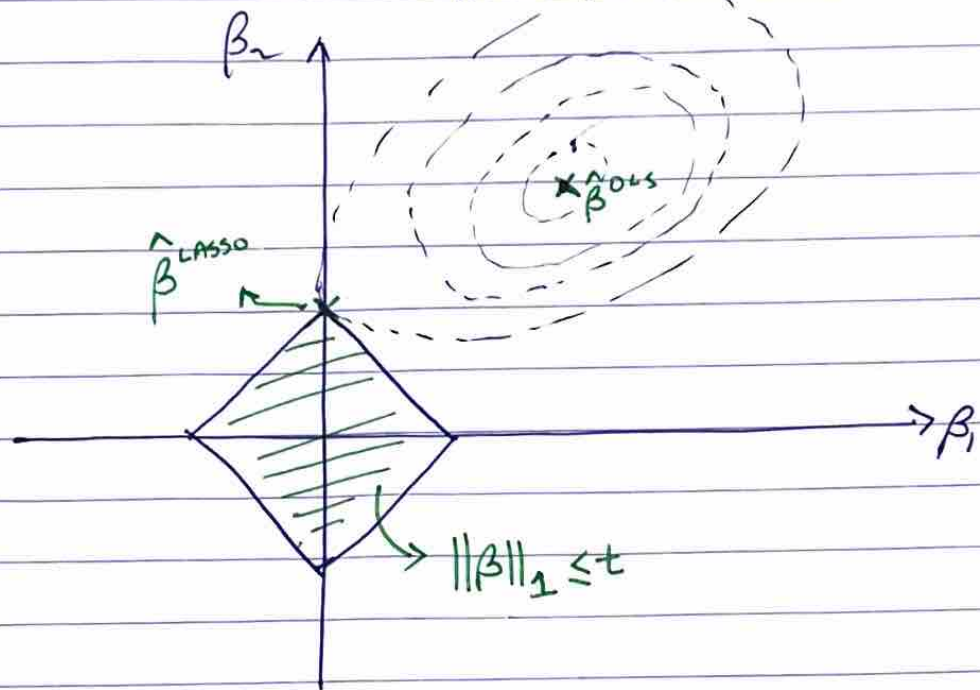
★ No closed form solⁿ of $\hat{\beta}^{\text{LASSO}}$; need to solve numerically

Q: Why ℓ_1 regularizer?

A: Because this is equivalent to solving a constrained optimization of the form:

$$\min_{\beta \in \mathbb{R}^p} \|Y - X\beta\|^2$$

$$\text{s.t. } \|\beta\|_1 \leq t$$



- ℓ_1 norm balls tend to be "spiky"

$\Rightarrow$ Solution tends to be at a corner,
which is sparse (i.e. several
entries zero)

Moral: ℓ_1 regularization encourages Sparsity
of ML model being fitted.